HARKIRAT SINGH

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala — Patiala, Punjab
Bachelor of Engineering in Computer Engineering(CGPA: 7.23/10) — 2023 – 2027

Khalsa College Public School, Amritsar — Amritsar, Punjab
Class 12 - CBSE (Percentage: 81.5%) — 2023

Holy Heart Presidency School, Amritsar — Amritsar, Punjab
Class 10 - ICSE (Percentage: 91.2%) — 2021

TECHNICAL SKILLS

- **Languages & Data:** Python, SQL, Java, Pandas, NumPy, Excel
- **Data Analytics:** Data Cleaning, Data Validation, Exploratory Data Analysis, Statistical Analysis, Data Interpretation
- **Business Intelligence:** Power BI, Tableau, Data Visualization, Dashboard Development, KPI Analysis
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Generative AI:** LangChain, Prompt Engineering, Vector Databases, RAG
- **Core CS:** Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Professional:** Analytical Problem Solving, Communication, Cross-Functional Collaboration, Leadership, Adaptability

PROJECTS

Enterprise Supply Chain & Inventory Analytics Dashboard — *Excel, SQL, Power BI, Python*

- Analyzed **180K+ orders** across **23 global regions** to identify delivery bottlenecks, SLA breaches, and fulfillment gaps.
- Developed **OTD, OTIF, SLA Breach, Perfect Order** and shipping-variance KPIs using structured data modeling.
- Identified **40.9% OTD, 57.3% SLA breach, and 14% Perfect Order Rate** to drive process improvements.
- Designed an interactive **Power BI dashboard** with regional drill-downs, SLA heatmaps, and **36-month trends**.

ChurnSight — Predictive Customer Attrition Platform — *Python, ML, Streamlit*

- Building an XGBoost forecasting and simulation model on 7,000+ subscriber records to predict churn probability and quantify LTV -at -risk, paralleling forecasting-to-business-outcome modeling used in analyst engagements.
- Applying SHAP -based explainability to convert model outputs into clear, business-meaningful drivers —translating technical analysis into executive communication.
- Used AI-assisted development tools (Copilot/Claude) throughout to accelerate coding, testing, debugging, and documentation workflows -reflecting the technology design, development, testing & implementation lifecycle.

SymptomScope AI — *AI/ML-Based Symptom Intelligence System* — Live Demo — Github

Next.js, TypeScript, Python, FastAPI, MongoDB, Clerk, LangChain, ChromaDB

- Built an AI symptom checker using Decision Tree, Random Forest & Naive Bayes for disease prediction, severity scoring.
- Engineered a Groq & LangChain RAG pipeline for structured, personalized recovery plans and medical guidance.
- Developed a goal-driven Groq AI agent with tool calling, context memory, and approval-based health actions.
- Built a RAG-based medical knowledge base using ChromaDB and TF-IDF embeddings for context-aware health responses.

LEADERSHIP & EXTRACURRICULARS

Joint Secretary– IETE Students Forum, TIET Patiala — May 2025 – May 2026

- Led and managed a 27-member cross-functional executive team to execute over 10 large-scale technical events and student initiatives for 450+ participants, streamlining budgets, operational timelines, and task allocations across multiple concurrent projects while leveraging data-driven engagement insights to boost membership by 40%.

Student Ambassador— LetsUpgrade EdTech Pvt. Ltd. — August 2024

- Coordinated targeted outreach initiatives to onboard 45+ students into skill-building workshops, analyzed engagement and referral metrics to optimize high-performing channels, and directly engaged prospective students to communicate relevant opportunities, ultimately improving referral-based campaign conversion rates by 20

CERTIFICATIONS

Certifications

- **TATA(Forage)** — Data Visualisation: Empowering Business with Effective Insights
- **BCG X(Forage)** — Introduction to Data for Decision Makers
- **TATA(Forage)** — GenAI Powered Data Analytics